A Versatile Probability Distribution Model for Wind Power Forecast Errors and Its Application in Economic Dispatch

Zhao-Sui Zhang, *Student Member, IEEE*, Yuan-Zhang Sun, *Senior Member, IEEE*, David Wenzhong Gao, *Senior Member, IEEE*, Jin Lin, *Student Member, IEEE*, and Lin Cheng, *Member, IEEE*

***Abstract*—The existence of wind power forecast errors is one of the most challenging issues for wind power system operation. It is difficult to find a reasonable method for the representation of forecast errors and apply it in scheduling. In this paper, a probability distribution model named "versatile distribution" is formulated and developed along with its properties and applications. The model can well represent forecast errors for all forecast timescales and magnitudes. The incorporation of the model in economic dispatch (ED) problems can simplify the wind-induced uncertainties via a few analytical terms in the problem formulation. The ED problem with wind power could hence be solved by the classical optimization methods, such as sequential linear programming which has been widely accepted by industry for solving ED problems. Discussions are also extended on the incorporation of the proposed versatile distribution into unit commitment problems. The results show that the new distribution is more effective than other commonly used distributions (i.e., Gaussian and Beta) with more accurate representation of forecast errors and better formulation and solution of ED problems.**

***Index Terms*—Cumulative distribution function (CDF), economic dispatch (ED), forecast, forecast span (FS), probability density function (PDF), sequential linear programming (SLP), versatile distribution, wind power.**

I. Introduction

WIND power still could not be forecasted with satisfactory accuracy under current technologies. The error from day-ahead point prediction (also called deterministic prediction) can be as high as 25% $\sim$ 40% [1] of the installed capacity of a wind farm. The forecast error can be even larger during times of rare events, such as during an overspeed shutdown event. Therefore, it is important to also provide end users with forecast uncertainties [2]. This is especially the case for some decision-making areas such as economic dispatch (ED), i.e., short-term decision of the optimal output of a number of units connected to the grid to meet the load at the lowest operating cost. To this end, there are primarily two crucial problems required to be addressed.

The first problem is how to well represent forecast errors of wind power. The forecast error is usually assumed as a random variable that obeys the law of Gaussian [3]–[5] or Beta [6], [7] distribution. However, many studies have demonstrated the lack of generalization of Gaussian and Beta distributions for some forecast timescales and magnitudes. For example, Hodge and Milligan [8] found that the shape of probability density function (PDF) of forecast error varies greatly with the length of forecast timescale. For minute-level forecast timescales (within 1 hour), the PDFs of the sample historical data of forecast errors have more pronounced peaks, steeper shoulders and fatter tails than those generated by Gaussian and Beta distributions using the same sample [8]. In addition, the PDF of forecast error is a conditional probability function with respect to the forecast value. When the forecast value is close to 0 or 1 p.u., a distortion appears in the PDF curve [9] because the wind power output should be bounded within 0 to 1 p.u. Gaussian distribution is not a good description of this distortion because its PDF curve is symmetrical. An alternative approach is to assume forecast errors not following any continuous distribution. Jabr and Pal [10] characterized the distribution of forecast errors precisely by a series of discrete bars, i.e., probability density histogram (PDH). However, it is time-consuming to apply this method since discrete computation techniques increase the number of computation variables, especially when high computation accuracy is required.

The second problem is how to mathematically manage the forecast error in ED problems. Numerous ED models incorporating uncertain wind power have been developed. For instance, Pappala *et al.* [11] proposed to model the wind power uncertainties using scenario generation and reduction technologies. Miranda and Hang [12] presented a fuzzy model to describe wind uncertainties and dispatcher's attitudes toward risk. Hetzer *et al.* [13] included the factors in an ED model to account for underestimation and overestimation of available wind power. Albadi and El-Saadany [14] represented the uncertainty associated with wind using a chance-constrained formulation. The inclusion of

Manuscript received July 09, 2012; revised November 19, 2012 and February 06, 2013; accepted February 21, 2013. Date of publication March 18, 2013; date of current version July 18, 2013. This work was supported by National High-Technology Research and Development Program ("863" Program) of China (2011AA05A103). Paper no. TPWRS-00793-2012.

Z.-S. Zhang is with the State Key Laboratory of Power Systems, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China. He is also with the Earth Institute, Columbia University, New York, NY 10025 USA (e-mail: zhangzhaosui@gmail.com).

Y.-Z. Sun, J. Lin, and L. Cheng are with the State Key Laboratory of Power Systems, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China (e-mail: yzsun@tsinghua.edu.cn; tulinjin@gmail.com; chenglin@tsinghua.edu.cn).

D. W. Gao is with the Department of Electrical and Computer Engineering, University of Denver, Denver, CO 80208 USA (e-mail: wenzhong.gao@du.edu).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TPWRS.2013.2249596

wind uncertainties in ED models increases the nonlinearity and complexity of ED problems. As a result, some classical algorithms, such as equal incremental cost criterion [15], sequential linear programming (SLP) [16] and Lagrange relaxation [17], can hardly be implemented and/or the computation time is greatly increased. Some heuristic algorithms, such as particle swarm algorithm [18], genetic algorithm [19] and simulated annealing algorithm [20], are deemed as the only option to solve the ED problem with wind power, although they do not always guarantee a globally optimal solution [21]. However, since the classical algorithms have been proven dependable by much successful operating experience of operators in the past, there still exists a pressing need to develop the ED algorithms to deal with wind in the classical fashion. The challenges are how to guarantee the feasibility and dependability of classical algorithms once wind power uncertainties are included in ED problems.

This paper aims to handle the above two problems. A probability distribution model, named "versatile distribution", is formulated to represent forecast errors. In the prior work [22], the versatile distribution was initially presented and it was proven more accurate to represent forecast errors than Gaussian and Beta distributions. In this work, the versatile distribution is further developed along with its properties and applications. Actually, it is easy to find or create a distribution model to represent forecast errors but hard to guarantee that this distribution is friendly to ED algorithms. For example, we can use a piecewise function to represent forecast error distribution, whereas it may complicate the solution procedures of ED problems. From the engineering standpoint, if the developed distribution model does not have good application prospects, its value decreases. Therefore, to highlight the value of the versatile distribution, much attention is paid to the development of its application on scheduling, which is also the main contribution of this paper.

In this paper, the versatile probability distribution model is firstly formulated and its mathematical properties are developed. Secondly, the qualitative and quantitative comparisons between the versatile distribution and other commonly used distributions (i.e., Gaussian and Beta) are studied, and the versatile distribution is proven better at representing forecast errors. Thirdly, the incorporation of the new distribution in a typical ED problem including wind power is investigated. By applying the versatile distribution, the ED problem with wind power can be solved by the classical algorithm based on SLP, which was not available in the past due to the introduction of wind uncertainties and the adoption of Gaussian and Beta distributions. Compared to using Gaussian and Beta distributions, the simulations show that the ED solution is more accurate and faster when using the versatile distribution. At last but not least, discussions are extended to the incorporation of versatile distribution in unit commitment (UC) problems.

II. Formulation of Versatile Distribution Model

The PDF and cumulative distribution function (CDF) of versatile distribution are as in (1) and (2) [22]:

$$f(x|\alpha,\beta,\gamma) = \frac{\alpha\beta e^{-\alpha(x-\gamma)}}{\left(1+e^{-\alpha(x-\gamma)}\right)^{\beta+1}} \tag{1}$$

$$F(x|\alpha,\beta,\gamma) = \left(1+e^{-\alpha(x-\gamma)}\right)^{-\beta} \tag{2}$$

where x is a random variable and α, β, γ are the shape parameters. The inverse function of CDF is as follows:

$$F^{-1}(c|\alpha,\beta,\gamma) = \gamma - \frac{1}{\alpha}\ln(c^{-1/\beta} - 1) \tag{3}$$

where c is the confidence level.

Two valuable mathematical properties of versatile distribution are summarized as follows.

- Property 1: The versatile distribution can well represent wind power forecast errors for any forecast timescale and forecast magnitude.
  The generality of versatile distribution mentioned in Property 1 is attributed to the adjustable shape parameters. Through adjusting the shape parameters to suitable values, the versatile distribution can better represent forecast errors compared with Gaussian and Beta distributions. This property will be further proven with numerical examples in Section III.
- Property 2: The CDF and its inverse function of versatile distribution have analytical forms as shown in (2) and (3), which can facilitate the algorithm of the ED problem considering wind power uncertainties.
  In the ED model with wind power, there usually exits chance constraints regarding available wind power [14], [23] and additional factors of objective function to account for underestimation and overestimation of available wind power [5], [10], [13]. A classical algorithm like SLP usually requires the constraints to be linearized and partial derivatives of objective function to be calculated first. Actually, the chance constraints and derivatives with respect to wind power are expressions related to CDF or its inverse function (see details in Section IV-B). The CDF and its inverse function can be analytically computed if the proposed versatile distribution is used, which apparently enhances the computation efficiency. In contrast, Gaussian and Beta distributions do not have such property as their CDFs must be obtained from numerical integration of PDF. Property 2 will be further proven in Section IV.

III. Representation of Wind Power Forecast Errors Using Versatile Distribution Model

In this section, we firstly introduce the method of generating the PDFs and CDFs of wind power output for a given forecast. Then the proposed versatile, Gaussian and Beta distributions are compared using qualitative and quantitative analysis to identify the best approximation for an actual distribution.

Notice that two definitions are easily confused: the one is the distribution of forecast error; the other one is the distribution of wind power output for a given forecast. As mentioned in Section I, the distribution of error/output is a conditional probability function with respect to the forecast value. For the same forecast value, the curve of output distribution is obtained by shifting that of forecast error distribution to the right by a magnitude of the corresponding forecast value. Since the two kinds of distributions have the same shape for the same forecast value,

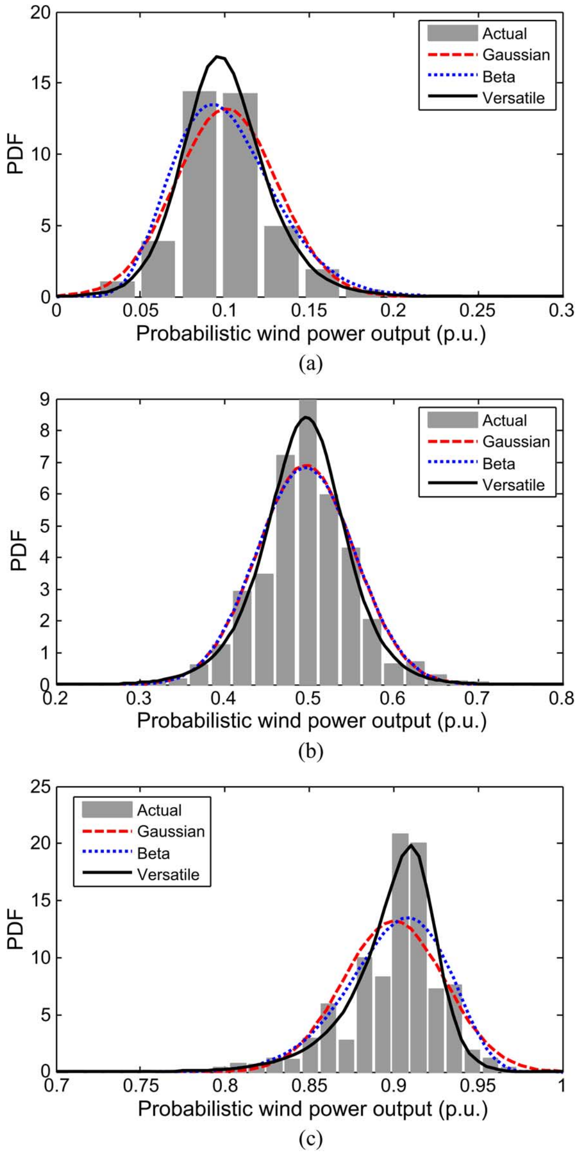


Fig. 1. Actual, Gaussian, Beta, and versatile PDFs of output for the forecast value located in FS m. The subfigures (a), (b), and (c) stand for three independent cases of $m = 3$, 13, and 23, respectively. Note that the parameters of versatile distribution are obtained by curve fitting the actual PDF. (a) $m = 3$; (b) $m = 13$; (c) $m = 23$.

they are essentially consistent with each other. The only difference between the two kinds of distributions is that they have different locations. The output distribution is located within an identical range of [0, 1] p.u. on the horizontal ordinate and its peak is around w_f, where w_f is assumed as the forecast value. By contrast, the forecast error distribution is located within a conditional range with respect to w_f, i.e., $[-w_f, 1 - w_f]$, and its peak is around 0. In order to better visually distinguish different distributions regarding different forecast conditions, the form of output distribution is used subsequently (e.g., Fig. 1).

A. Generation of PDF and CDF of Output for a Given Forecast

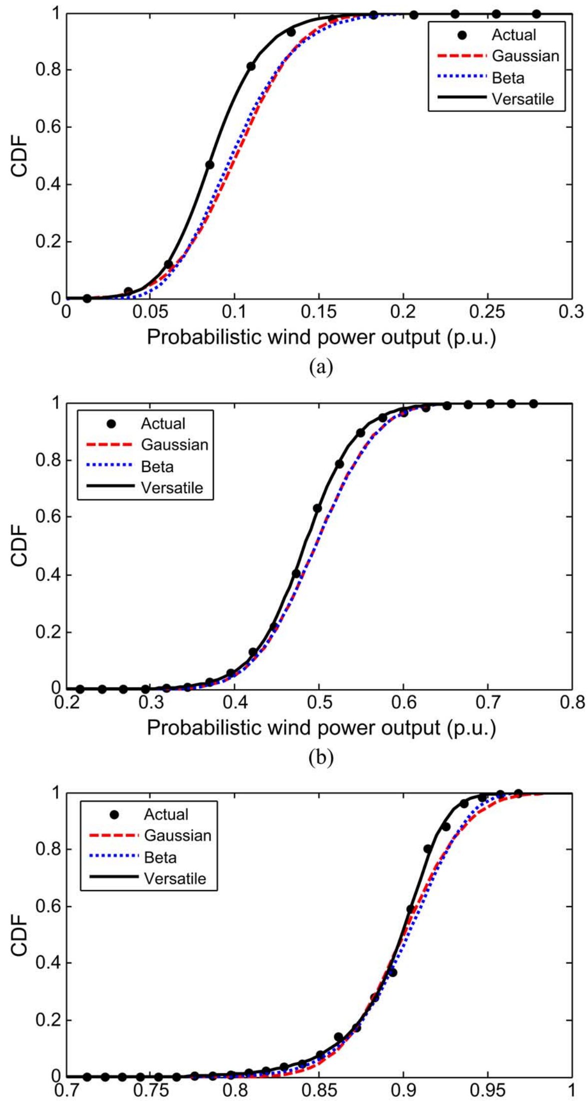
The 5-min operational data of actual output in 2008 are collected from a 150-MW wind power plant located in Tongliao City, Inner Mongolia, China. The corresponding time series of 10-min-ahead forecasts are generated using the persistence forecast model [8], [22], [24]. All the data are standardized in per unit system with the installed capacity (150 MW) as the base value.

Before the generation of PDF/CDF, we firstly define an important terminology—**forecast span (FS)**. The forecast range (i.e., [0, 1] p.u.) is divided into M even spans (i.e., $[0, 1/M]$, $[1/M, 2/M], \ldots, [(M-1)/M, 1]$) and the width of each span is $1/M$. Each span is called an "FS" in this paper. FS m $(m = 1, 2, \ldots, M)$ corresponds to the range of $[(m-1)/M, m/M]$. The time series data of forecasts and actual outputs are one-to-one pairs at each time instant. For each forecast value located within FS m, there is an associated output. Thus FS m corresponds to a group of associated outputs, denoted by "output dataset for FS m". M is a user-defined integer according to the scale of historical data [22], [25]. To guarantee sufficient data points in the output dataset for FS m, a value of $M = 25$ is found to work well for the whole year operational data used here.

The generation procedure for the PDFs is listed as follows [22], [24], [25]: 1) The mth actual PDF is represented by the PDH (see the discrete bars in Fig. 1) generated from the statistical analysis of the output dataset for FS m; 2) The parameters of the mth Gaussian PDF (see the red dashed line in Fig. 1) are equal to the mean and variance of the output dataset for FS m; 3) The parameters of the mth Beta PDF (see the blue dotted line in Fig. 1) are related to the mean and variance of the output dataset for FS m and the relationships are given by study [26]; 4) The parameters of the mth versatile PDF (see the black solid line in Fig. 1) are weakly correlated with the mean and variance, but determined by curve fitting for the mth actual PDF based on nonlinear least squares estimation. In Fig. 1, for each case of a specific "m", the actual, Gaussian, Beta and versatile PDFs are compared, and three independent cases for $m = 3$, 13, 23 are plotted separately on Fig. 1(a)–(c). Note that the PDF curve remains the same for the forecast values located in the same FS, whereas it varies with the FS (see Fig. 1). This means that 25 PDFs are needed for the 25 FSs.

The actual CDF (see the discrete dots in Fig. 2) is represented by a group of discrete cumulative probabilities computed from the PDH. The Gaussian and Beta CDFs (see the red dashed and blue dotted lines in Fig. 2) are obtained from numerical integration of Gaussian and Beta PDFs, respectively. The versatile CDF (see the black solid line in Fig. 2) is obtained from curve fitting the actual CDF using nonlinear least squares estimation.

It must be emphasized that, different from Gaussian and Beta distributions, the shape parameters of versatile distribution generated by fitting the actual PDF (in Fig. 1) differs from those generated by fitting the actual CDF (in Fig. 2). The solutions obtained from the fitting are not unique and are influenced by the setting of initial values for the parameters. Thus, the values of parameters generated by PDF fitting are sometimes different from those generated by CDF fitting. In fact, CDF fitting is preferable because the wind-induced uncertain items, i.e., chance constraints and derivatives of objective function, in an ED problem are related to CDF (see details in Section IV-B). Thus, CDF fitting method will be adopted in the subsequent text.



(a) (b) (c)

Fig. 2. Actual, Gaussian, Beta, and versatile CDFs of output for the forecast value located in FS m. The subfigures (a), (b), and (c) stand for three independent cases of $m = 3$, 13, and 23, respectively. Note that the parameters of versatile distribution are obtained by curve fitting the actual CDF. (a) $m = 3$; (b) $m = 13$; (c) $m = 23$.

We can generate a look-up table of the shape parameters of versatile distribution for each FS (e.g., Table I) based on CDF fitting method. For different wind power plants or different forecast timescales (10 min in this study), the corresponding look-up tables are different.

As mentioned in this section, the solutions obtained from the curve fitting (using the function "nlinfit" in Matlab) depend on the setting of initial values for the shape parameters. If the initial values are chosen arbitrarily, some solutions for the parameters in Table I may have different signs or orders of magnitude compared to the remaining ones. However, the 1st γ value, though with an opposite sign compared with others, is actually not a discrepancy. The reason is that there appears to be an approximately linear relationship between the γ value and the FS number m, and the 1st γ value also complies with this relationship. In fact, as long as the versatile distribution represents the actual distribution better than Gaussian and Beta distributions, the generated parameters are acceptable, even though there seem to be "discrepancy" on the values of some of them.

TABLE I
LOOK-UP TABLE OF THE SHAPE PARAMETERS OF VERSATILE DISTRIBUTION FOR THE WIND POWER PLANT WITH 10-MIN-AHEAD PREDICTION THROUGH CURVE FITTING THE ACTUAL CDF

FS No. m	Forecast value range (p.u)	Shape parameters (p.u.) α	β	γ
1	[0.00, 0.04]	65.73	2.07	-0.02
2	[0.04, 0.08]	61.78	2.21	0.03
3	[0.08, 0.12]	58.05	1.89	0.07
4	[0.12, 0.16]	49.70	1.60	0.11
5	[0.16, 0.20]	41.85	1.98	0.15
6	[0.20, 0.24]	38.75	1.80	0.19
7	[0.24, 0.28]	40.50	1.09	0.24
8	[0.28, 0.32]	38.58	1.13	0.28
9	[0.32, 0.36]	33.55	1.50	0.30
10	[0.36, 0.40]	33.76	1.22	0.36
11	[0.40, 0.44]	33.92	1.26	0.39
12	[0.44, 0.48]	33.26	1.11	0.44
13	[0.48, 0.52]	32.96	0.98	0.48
14	[0.52, 0.56]	30.83	1.06	0.52
15	[0.56, 0.60]	32.80	0.91	0.56
16	[0.60, 0.64]	36.65	0.74	0.62
17	[0.64, 0.68]	32.63	1.08	0.64
18	[0.68, 0.72]	40.64	0.76	0.69
19	[0.72, 0.76]	38.41	0.71	0.73
20	[0.76, 0.80]	40.40	0.81	0.77
21	[0.80, 0.84]	37.37	0.82	0.81
22	[0.84, 0.88]	63.62	0.41	0.88
23	[0.88, 0.92]	115.35	0.34	0.92
24	[0.92, 0.96]	97.33	0.70	0.93
25	[0.96, 1.00]	153.47	0.41	0.97

B. *Qualitative Comparison Among Possible Distributions*

Fig. 1 compares the actual (discrete bars), Gaussian (red dashed line), Beta (blue dotted line) and versatile (black solid line) PDFs of wind power output for the forecast value located in FS m. Three independent FSs ($m = 3$, 13, 23) are studied here. As shown in Fig. 1, for such short forecast timescale of 10 min, the pronounced peak of the actual PDF cannot be properly represented by Gaussian and Beta distributions. For the case of $m = 3$ [see Fig. 1(a)] or 23 [see Fig. 1(c)], the actual PDF skews a bit towards one side, whereas the Gaussian PDF does not skew at all. Compared with Gaussian and Beta distributions, the versatile distribution better models the actual distribution for any case.

In Fig. 2, the three cases of $m = 3$, 13, 23 are compared regarding the CDFs. As shown in Fig. 2, the Gaussian and Beta CDFs deviate far away from the actual CDF while the versatile distribution again exhibits a better matching for any case. The results can be explained as follows. As mentioned in Section III-A, the Gaussian and Beta CDFs are derived by numerical integration of PDFs. Since the estimation error in Gaussian/Beta PDF is further propagated and amplified in the process of numerical integration to obtain a CDF, the Gaussian/Beta CDF involves an accumulated error. By contrast, the versatile CDF is directly generated by curve fitting the actual CDF, which does not result in such accumulated error.

TABLE II
RMSEs (CDF) FOR DIFFERENT DISTRIBUTIONS FOR THE WIND POWER PLANT WITH 10-MIN-AHEAD PREDICTION

FS No. m	Forecast value range (p.u.)	RMSEs Gaussian	Beta	Versatile
1	[0.00, 0.04]	0.070	0.033	0.001
2	[0.04, 0.08]	0.071	0.064	0.002
3	[0.08, 0.12]	0.056	0.050	0.004
4	[0.12, 0.16]	0.056	0.051	0.001
5	[0.16, 0.20]	0.057	0.052	0.006
6	[0.20, 0.24]	0.042	0.038	0.003
7	[0.24, 0.28]	0.047	0.044	0.006
8	[0.28, 0.32]	0.043	0.040	0.006
9	[0.32, 0.36]	0.050	0.048	0.005
10	[0.36, 0.40]	0.042	0.041	0.002
11	[0.40, 0.44]	0.046	0.045	0.006
12	[0.44, 0.48]	0.044	0.044	0.007
13	[0.48, 0.52]	0.039	0.039	0.006
14	[0.52, 0.56]	0.039	0.039	0.003
15	[0.56, 0.60]	0.040	0.041	0.002
16	[0.60, 0.64]	0.031	0.033	0.008
17	[0.64, 0.68]	0.035	0.038	0.004
18	[0.68, 0.72]	0.034	0.037	0.006
19	[0.72, 0.76]	0.033	0.037	0.008
20	[0.76, 0.80]	0.030	0.036	0.004
21	[0.80, 0.84]	0.028	0.034	0.010
22	[0.84, 0.88]	0.032	0.034	0.007
23	[0.88, 0.92]	0.040	0.042	0.012
24	[0.92, 0.96]	0.040	0.048	0.012
25	[0.96, 1.00]	0.040	0.042	0.028

C. Quantitative Comparison Among Possible Distributions

To quantitatively compare the possible distributions, the root-mean-square error (RMSE) [22] between the actual CDF and the versatile/Gaussian/Beta CDF is used as shown in (4):

$$\mathrm{RMSE}_m = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(F_{act,m}(x_n) - F_{sim,m}(x_n)\right)^2} \tag{4}$$

where subscript m is the serial number of FS, x_n is the wind power output corresponding to the nth bar of the actual PDF as shown in Fig. 1, N is the total number of actual PDF bars, $F_{act,m}(x_n)$ is the actual CDF when the wind power output is no more than x_n, and $F_{sim,m}(x_n)$ is the versatile/Gaussian/Beta CDF when the wind power output is no more than x_n.

Table II compares the RMSEs of different distributions and the RMSE of versatile distribution (in shadow) is much smaller than that of Gaussian/Beta distribution for each FS. Therefore, the versatile distribution can best approximate the actual distribution among the three distributions.

IV. Application of Versatile Distribution Model in Economic Dispatch Problems

A. Basic Economic Dispatch Model Used

For the sake of emphasizing the method for modeling wind power uncertainties, this study simplifies the ED model by neglecting ramping rate limits, transmission constraints and transmission losses. This is acceptable because the ramping rate constraint is linear, and the transmission constraint and power balance constraint considering losses can be linearized using DC load flow models. Linear constraints do not affect the effectiveness of SLP-based algorithms. Moreover, the simplified ED model is especially practicable to real-time ED problems where the dispatch interval is short (in minute level) and the decision is made for just the next single interval.

A typical ED model with wind power is used here as shown in (5)–(12) [13]. Equation (5) is the objective function, (6) is the power balance constraint, (7) and (8) are the power generation limit constraints, and (9)–(12) are the regulation reserve constraints:

$$\begin{aligned}\min \sum_{i=1}^{I} C_i(p_i) &+ \sum_{j=1}^{J} C_{w,j}(w_j)\\ &+ \sum_{j=1}^{J} C_{p,j}(w_{av,j} - w_j)\\ &+ \sum_{j=1}^{J} C_{r,j}(w_j - w_{av,j})\end{aligned} \tag{5}$$

s.t.

$$\sum_{i=1}^{I} p_i + \sum_{j=1}^{J} w_j = L \tag{6}$$

$$p_{\min,i} \le p_i \le p_{\max,i} \tag{7}$$

$$0 \le w_j \le w_{r,j} \tag{8}$$

$$0 \le r_{u,i} \le \min\{p_{\max,i} - p_i, r_{u,\max,i}\} \tag{9}$$

$$0 \le r_{d,i} \le \min\{p_i - p_{\min,i}, r_{d,\max,i}\} \tag{10}$$

$$\sum_{i=1}^{I} r_{u,i} \ge \sum_{j=1}^{J}(w_j - w_{av,j}) \tag{11}$$

$$\sum_{i=1}^{I} r_{d,i} \ge \sum_{j=1}^{J}(w_{av,j} - w_j) \tag{12}$$

where I is the total number of conventional power plants (CPPs), J is the total number of wind power plants (WPPs), C_i is the cost function of CPP i, $C_{w,j}$ is the cost function of WPP j, $C_{p,j}$ is the underestimation cost function for not using all available wind power of WPP j, $C_{r,j}$ is the overestimation cost function for failing to produce the scheduled wind power of WPP j at the scheduled time, p_i is the scheduled power of CPP i, $p_{\min,i}$ and $p_{\max,i}$ are the lower and upper output limits of CPP i, w_j is the scheduled power of WPP j, $w_{av,j}$ is the actual available power of WPP j, $w_{r,j}$ is the installed capacity of WPP j, L is the system demand which is deemed as constant, $r_{u,i}$ and $r_{d,i}$ are the available amount of up and down regulation reserves provided by CPP i, and $r_{u,\max,i}$ and $r_{d,\max,i}$ are the maximum amount of up and down regulation reserves that CPP i is capable of providing within a specific time period (e.g., 10 min). The first two items of the objective function are detailed as in (13) and (14):

$$C_i(p_i) = a_i p_i^2 + b_i p_i + c_i \tag{13}$$

$$C_{w,j}(w_j) = d_j w_j \tag{14}$$

where a_i, b_i, c_i are the fuel cost coefficients of CPP i and d_j is the direct cost coefficient of WPP j.

Because the available wind power $(w_{av,j})$ is a random variable which can hardly be predicted accurately, how to handle the uncertain items consisting of $w_{av,j}$, i.e., (5), (11) and (12), in an appropriate way is the main challenge for the ED solution. Currently, there are two popular approaches that can be used: One expresses the probabilistic underestimation and overestimation costs ($C_{p,j}$ and $C_{r,j}$) as the forms of expectation values as shown in (15) and (16) [5], [10], [13]; The other approach replaces the probabilistic constraints of (11) and (12) by the chance-constrained formulas of (17) and (18) [14], [23]:

$$\begin{aligned} C_{p,j}(w_{av,j}-w_j) &= k_p(w_{av,j}-w_j) \\ &= k_p\int_{w_j}^{w_{r,j}}(x-w_j)g_{j,m}(x)dx \end{aligned} \tag{15}$$

$$\begin{aligned} C_{r,j}(w_j-w_{av,j}) &= k_r(w_j-w_{av,j}) \\ &= k_r\int_{0}^{w_j}(w_j-x)g_{j,m}(x)dx \end{aligned} \tag{16}$$

$$\Pr\left\{\sum_{i=1}^{I} r_{u,i} \geq \sum_{j=1}^{J}(w_j-w_{av,j})\right\} \geq, c_u \tag{17}$$

$$\Pr\left\{\sum_{i=1}^{I} r_{d,i} \geq \sum_{j=1}^{J}(w_{av,j}-w_j)\right\} \geq c_d \tag{18}$$

where k_p and k_r are the underestimation and overestimation cost coefficients, $g_{j,m}(\cdot)$ is the PDF of the output of WPP j for the forecast value located in FS m, and c_u and c_d are the confidence levels for having sufficient up and down regulation reserves. The underestimation cost is now equivalent to the expected cost of wind power surplus as shown in (15). The overestimation cost is equivalent to the expected cost of wind power deficit as shown in (16). Currently, some electricity markets have stipulated that wind generation deviations from original schedules are penalized [27]. This deviation penalty can be well described by (15) and (16). Equations (17) and (18) imply that the probabilities of (11) and (12) being satisfied should be greater than the confidence levels of c_u and c_d, respectively.

The basic ED model used in this paper is the stochastic programming problem described by (5)–(10) and (13)–(18).

B. Challenge for SLP-Based Algorithms

One of the most important procedures of SLP is to approximate the objective function by a Taylor polynomial in the vicinity of an operating point. That is, the partial derivatives of objective function must be firstly computed. The derivatives of items (15) and (16) of objective function with respect to the scheduled wind power are given by

$$\begin{aligned} \frac{\partial C_{p,j}}{\partial w_j} &= -k_p\int_{w_j}^{w_{r,j}} g_{j,m}(x)dx \\ &= k_p\cdot(G_{j,m}(w_j)-G_{j,m}(w_{r,j})) \end{aligned} \tag{19}$$

$$\begin{aligned} \frac{\partial C_{r,j}}{\partial w_j} &= k_r\int_{0}^{w_j} g_{j,m}(x)dx \\ &= k_r\cdot(G_{j,m}(w_j)-G_{j,m}(0)) \end{aligned} \tag{20}$$

where $G_{j,m}(\cdot)$ is the CDF of the output of WPP j for the forecast value located in FS m. As addressed in Section II, if $G_{j,m}$ is approximated by Gaussian or Beta CDF, it does not have analytical form and can only be obtained by numerical integration of PDF. However, numerical integration increases computation time, and also results in an accumulated error in CDF according to the analysis in Section III-B.

Another important procedure of SLP is to convert the chance constraints (17) and (18) to linear and deterministic ones. Equations (17) and (18) can be deduced to alternative expressions as follows:

$$\begin{aligned} &\left\{\begin{aligned} &\Pr\left\{\sum_{i=1}^{I} r_{u,i} \geq \sum_{j=1}^{J}(w_j-w_{av,j})\right\} \\ &\quad = 1-\Pr\left\{\sum_{j=1}^{J} w_{av,j} \leq \sum_{j=1}^{J} w_j - \sum_{i=1}^{I} r_{u,i}\right\} \\ &\quad = 1-G_\Sigma\left(\sum_{j=1}^{J} w_j - \sum_{i=1}^{I} r_{u,i}\right) \\ &\Pr\left\{\sum_{i=1}^{I} r_{u,i} \geq \sum_{j=1}^{J}(w_j-w_{av,j})\right\} \geq c_u \end{aligned}\right. \\ &\Rightarrow \sum_{j=1}^{J} w_j - \sum_{i=1}^{I} r_{u,i} \leq G_\Sigma^{-1}(1-c_u) \end{aligned} \tag{21}$$

$$\begin{aligned} &\left\{\begin{aligned} &\Pr\left\{\sum_{i=1}^{I} r_{d,i} \geq \sum_{j=1}^{J}(w_{av,j}-w_j)\right\} \\ &\quad = \Pr\left\{\sum_{j=1}^{J} w_{av,j} \leq \sum_{j=1}^{J} w_j + \sum_{i=1}^{I} r_{d,i}\right\} \\ &\quad = G_\Sigma\left(\sum_{j=1}^{J} w_j + \sum_{i=1}^{I} r_{d,i}\right) \\ &\Pr\left\{\sum_{i=1}^{I} r_{d,i} \geq \sum_{j=1}^{J}(w_{av,j}-w_j)\right\} \geq c_d \end{aligned}\right. \\ &\Rightarrow \sum_{j=1}^{J} w_j + \sum_{i=1}^{I} r_{d,i} \geq G_\Sigma^{-1}(c_d) \end{aligned} \tag{22}$$

where $G_\Sigma(\cdot)$ is the aggregated CDF of sum of the outputs from all the WPPs. As addressed in Section II, if G_Σ is approximated by Gaussian or Beta CDF, the inverse function G_Σ^{-1} does not have analytical form. For Gaussian or Beta distribution, the value of inverse CDF must be obtained by numerical search techniques. However, numerical search is time-consuming because every estimated CDF in a search loop is calculated using numerical integration.

Due to the above two problems, it is difficult to solve the ED problem with wind power by the classical algorithm based on SLP. The bottleneck is that there are no analytical mathematical expressions for the CDF and inverse CDF of Gaussian and Beta distributions.

C. Proposed SLP-Based Algorithm for the Economic Dispatch Problem Incorporating Wind Power

In this work, the versatile distribution model (1) and (2) is implemented in the ED model described by (5)–(10) and (13)–(18), so (19)–(22) can be simplified as in the following analytical forms:

$$\frac{\partial C_{p,j}}{\partial w_j} = k_p \cdot (F_{j,m}(w_j) - F_{j,m}(w_{r,j})) \tag{23}$$

$$\frac{\partial C_{r,j}}{\partial w_j} = k_r \cdot (F_{j,m}(w_j) - F_{j,m}(0)) \tag{24}$$

$$\sum_{j=1}^{J} w_j - \sum_{i=1}^{I} r_{u,i} \le F_{\Sigma}^{-1}(1-c_u) \tag{25}$$

$$\sum_{j=1}^{J} w_j + \sum_{i=1}^{I} r_{d,i} \ge F_{\Sigma}^{-1}(c_d) \tag{26}$$

where $F_{j,m}(\cdot)$ is the versatile CDF of the output of WPP j for the forecast value located in FS m, and $F_{\Sigma}(\cdot)$ is the aggregated versatile CDF of sum of the outputs from all the WPPs. According to Property 2 proposed in Section II, the CDF and inverse CDF of versatile distribution have analytical forms as shown in (2) and (3). This implies that the derivatives of (15) and (16) can be analytically computed by (23) and (24), and the chance constraints (17) and (18) are converted to the linear inequalities of (25) and (26). As a result, the ED problem considering wind power uncertainties can now be solved by SLP-based algorithm. The solution procedure of SLP in this paper is introduced as follows [16].

Near the operating point

$$(w_{1(k)}, w_{2(k)}, \ldots, w_{J(k)}, p_{1(k)}, p_{2(k)}, \ldots, p_{I(k)})$$

in iteration k, the objective function (5) can be analytically linearized by Taylor series expansion by using (23) and (24). Moreover, the nonlinear constraints (17) and (18) can be substituted by the linear constraints (25) and (26). Thereby, for the kth iteration, the stochastic nonlinear ED problem described by (5)–(10) and (13)–(18) can be simplified as a deterministic linear problem as shown in (27):

$$\begin{cases} \min\ H\Delta L + \sum_{i=1}^{I} h_{i(k)}(p_i - p_{i(k)}) + \sum_{j=1}^{J} h_{w,j(k)}(w_j - w_{j(k)}) \\ s.t.\ \sum_{i=1}^{I} p_i + \sum_{j=1}^{J} w_j = L - \Delta L \\ p_{\min,i} \le p_i \le p_{\max,i} \\ 0 \le w_j \le w_{r,j} \\ 0 \le \Delta L \le L \\ 0 \le r_{u,i} \le \min\{p_{\max,i} - p_i, r_{u,\max,i}\} \\ 0 \le r_{d,i} \le \min\{p_i - p_{\min,i}, r_{d,\max,i}\} \\ \sum_{j=1}^{J} w_j - \sum_{i=1}^{I} r_{u,i} \le F_{\Sigma}^{-1}(1-c_u) \\ \sum_{j=1}^{J} w_j + \sum_{i=1}^{I} r_{d,i} \ge F_{\Sigma}^{-1}(c_d) \\ |p_i - p_{i(k)}| \le \Delta p_{i(k)} \\ |w_j - w_{j(k)}| \le \Delta w_{j(k)} \end{cases} \tag{27}$$

where H is the user-defined penalty factor satisfying $H \gg \max\{h_{i(k)}, h_{w,j(k)}\}$, ΔL is the load shedding amount, $\Delta p_{i(k)}$ is the user-defined radius of the neighborhood of $p_{i(k)}$, $\Delta w_{j(k)}$ is the user-defined radius of the neighborhood of $w_{j(k)}$. Deducing from (13), (14), (23) and (24), the constant coefficients $h_{i(k)}$ and $h_{w,j(k)}$ appearing in model (27) are derived as shown in (28):

$$\begin{cases} h_{i(k)} = \left.\frac{\partial C_i}{\partial p_i}\right|_{p_i = p_{i(k)}} = 2a_i p_{i(k)} + b_i \\ h_{w,j(k)} = \left.\left(\frac{\partial C_{w,j}}{\partial w_j} + \frac{\partial C_{p,j}}{\partial w_j} + \frac{\partial C_{r,j}}{\partial w_j}\right)\right|_{w_j = w_{j(k)}} \\ = d_j - k_p F_{j,m}(w_{r,j}) - k_r F_{j,m}(0) + (k_p + k_r) F_{j,m}\left(w_{j(k)}\right). \end{cases} \tag{28}$$

Notice that the objective function in model (27) can be further simplified to (29) since a constant item does not affect the minimization of the objective function:

$$\min\ H\Delta L + \sum_{i=1}^{I} h_{i(k)} p_i + \sum_{j=1}^{J} h_{w,j(k)} w_j. \tag{29}$$

In the kth iteration, one can solve problem (27) using linear programming techniques, such as interior point algorithm and simplex algorithm [15], and obtain a solution of $(w_{1(k+1)}, w_{2(k+1)}, \ldots, w_{J(k+1)}, p_{1(k+1)}, p_{2(k+1)}, \ldots, p_{I(k+1)})$. In the $(k+1)$th iteration, the ED problem is linearized near the new operating point

$$(w_{1(k+1)}, w_{2(k+1)}, \ldots, w_{J(k+1)}, p_{1(k+1)}, p_{2(k+1)}, \ldots, p_{I(k+1)})$$

and is solved again. The iteration is repeated until the difference between the solution of the kth iteration and that of the $(k-1)$th iteration satisfies the convergence criterion, i.e., $|(w_{1(k+1)}, w_{2(k+1)}, \ldots, w_{J(k+1)}, p_{1(k+1)}, p_{2(k+1)}, \ldots, p_{I(k+1)}) - (w_{1(k)}, w_{2(k)}, \ldots, w_{J(k)}, p_{1(k)}, p_{2(k)}, \ldots, p_{I(k)})| < \varepsilon$, where ε is the convergence criterion parameter.

Note that the existence of feasible solution in each iteration loop is usually hard to predict. Thus, this work involves a load shedding variable ΔL in model (27) to ensure the existence of solution during the SLP iteration process. Moreover, to prevent the final solution from containing a non-zero load shedding, a penalty term $H \cdot \Delta L$ is added to the objective function as shown in (27). Thereby, as long as the original ED problem has a solution, non-zero load shedding merely appears in the iteration process occasionally and the true optimal solution with zero load shedding can be finally found.

D. Framework of the Incorporation of Versatile Distribution Model in Economic Dispatch Systems

Fig. 3 shows the framework of the ED system including wind power with the application of versatile distribution model. The

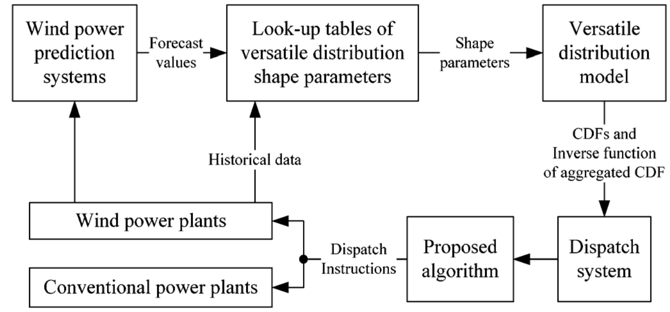


Fig. 3. Framework of the economic dispatch system incorporating wind power with the application of versatile distribution model.

TABLE III
PARAMETERS OF CONVENTIONAL POWER PLANTS

Gen. No.	Fuel cost coefficients ($/h)			Output limits (p.u.)		Reserve limits (p.u.)	
	a_i	b_i	c_i	$p_{min,i}$	$p_{max,i}$	$r_{u,\max,i}$	$r_{d,\max,i}$
1	100	200	10	0.4	1	0.2	0.2
2	120	150	10	0.4	1	0.2	0.2
3	40	180	20	0.4	1	0.2	0.2
4	60	100	10	0.4	1	0.2	0.2
5	40	180	20	0.4	1	0.2	0.2
6	40	180	20	0.4	1	0.2	0.2

look-up tables of versatile distribution parameters are generated by CDF fitting method using the historical forecast and output data from each WPP and their aggregation. Note that we always take the whole historical dataset available. The dataset should be supplemented by the latest data regularly (e.g., each day or each week) for the sake of improving accuracy. For a forecast value offered by the prediction system, the associated parameters of versatile distribution are sought out from the look-up table. The versatile CDF of output for the offered forecast and the inverse function of the aggregated versatile CDF are then provided to the dispatch system. Using the proposed SLP-based algorithm, the dispatch instructions of scheduled powers are computed and sent out to the generating units.

Notice that the generation of look-up table does not affect the computation efficiency of real-time ED. One reason is that the generation procedure only takes less than a second in the Matlab environment. Another reason is that, as mentioned above, the table only needs to be updated once a day or once a week.

E. Case Study

In order to verify the effectiveness of the proposed algorithm, the IEEE-30 bus system [28] with 6 CPPs and 1 WPP is studied. The parameters of CPPs are shown in Table III [28]. The WPP studied here is the same as the WPP aforementioned in Section III but with a different capacity of 100 MW. Every generation variable is expressed based on per unit system with 100 MW as the base value. The direct cost coefficient of wind power is 2 $/MWh, and the underestimation and overestimation cost coefficients k_p and k_r are 1.5 $/MWh and 3 $/MWh, respectively [29]. The confidence levels c_u and c_d are both 0.95, the system load L is 5 p.u. and the penalty factor H is 1 000 000 $/h. The computation is run in MATLAB R2012a on a Core-i7 2.70-GHz notebook computer.

The wind power forecast values varying from 0.1 to 0.9 p.u. with a step size of 0.1 p.u. are separately tested for the ED system. The proposed SLP-based algorithm is applied for each forecast value case. As a comparison, the SLP-based algorithms using the actual, Gaussian and Beta distributions are also studied, but based on discrete/numerical computation.

Fig. 4 shows the solutions of wind power schedules for each forecast value case. The solutions obtained using the versatile distribution are almost the same as those obtained using the actual distribution. In other words, the versatile distribution nearly does not sacrifice the accuracy of ED algorithms. In contrast, the solutions obtained using Gaussian and Beta distributions have

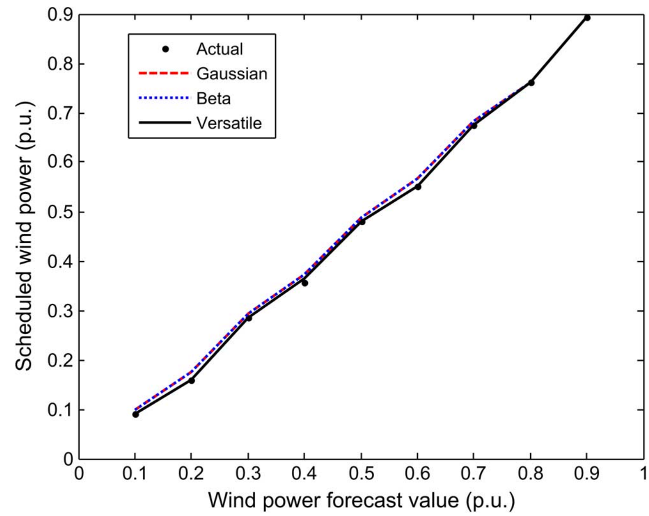


Fig. 4. Scheduled wind power versus wind power forecast.

TABLE IV
COMPUTATION TIME FOR DIFFERENT FORECAST CASES AND DISTRIBUTIONS

Forecast (p.u.)	Computation time (s) Actual	Gaussian	Beta	Versatile
0.1	1.66	0.96	1.40	0.96
0.2	1.65	1.10	1.83	1.07
0.3	1.59	0.92	1.98	0.88
0.4	1.77	1.21	2.52	1.06
0.5	1.67	1.02	2.77	0.96
0.6	1.70	0.95	2.96	1.01
0.7	1.68	1.01	3.35	0.98
0.8	1.80	1.18	3.91	1.08
0.9	1.58	0.89	3.39	0.89
Mean	1.68	1.03	2.68	0.99

errors. Similar results can be found from Fig. 5 except its subfigures regarding CPP 1 and 4. As the forecast value changes, the scheduled power of CPP 1 (p_1) maintains at its lower limit (0.4 p.u.) while that of CPP 4 (p_4) keeps at its upper limit (1 p.u.). By comparing the fuel cost coefficients of each unit (cf. Table III), one can find that CPP 1 is the most costly unit and CPP 4 is the cheapest one. Therefore, CPP 4 has the highest priority of dispatch and full generation is allowed as shown in Fig. 5. Meanwhile, CPP 1 has the lowest priority of dispatch and it is curtailed at the lower generation limit. From Figs. 4 and 5, it can be seen that the advantage of accuracy improvement using the versatile distribution seems not that obvious. However, note that the tested forecast timescale is just 10 min; and the versatile distribution will show a greater value if it is put into use for a longer time duration consisting of many 10-min segments. Taking the 0.2 p.u. forecast value case as an example, using the versatile distribution results in a cost saving of $1.04 in total compared with using Gaussian distribution within 10 min, thereby we can save $149.76 each day and save $54662.4 each year. It should be also noted that the tested system is only a small system with seven generating units. For a larger scale power system with dozens or hundreds of generating units, the total cost saving as a result of using the versatile distribution will be even larger.

Table IV compares the computation time of the algorithms using the actual, Gaussian, Beta and versatile distributions in

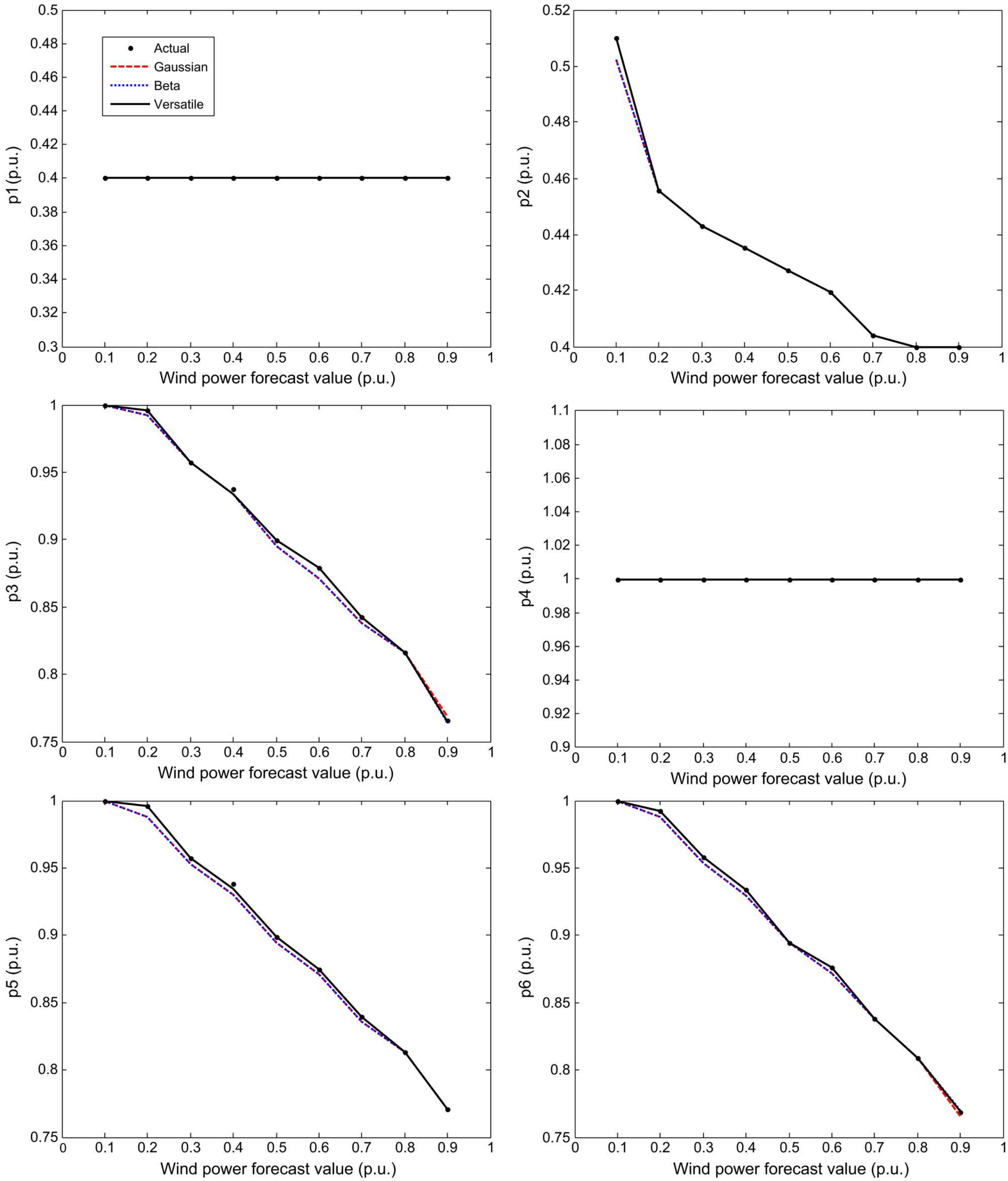


Fig. 5. Scheduled power from each conventional power plant versus wind power forecast.

each forecast value case. The proposed algorithm with the versatile distribution applied only spends an average computation time of 0.99 s, which is apparently the fastest one. Using the versatile distribution is particularly more effective than using Gaussian and Beta for low and high forecast values. It should be emphasized that the difference of mean computation time between using the versatile and Gaussian distributions is only 0.04 s. However, since the computation time using Gaussian distribution is a relatively small amount of 1.03 s, an improvement of 0.04 s means an increase of about 4% of the total computation time, which is actually considerable. Because the tested system is small, the ED computation is not that time-consuming in this case. If a larger scale power system with dozens or hundreds of generating units is considered, the total computation duration will become relatively long, so the improvement on computation speed using the versatile distribution may be even more considerable. Moreover, the static ED problem is generally less time-consuming because the decision is made for only the next single time interval. By contrast, for a dynamic ED problem containing constraints and decision variables coupling different time periods, the computation time is relatively long, so the absolute reduced computation time using the versatile distribution will definitely be much more significant.

V. Extension in a Unit Commitment Problem

The scheduling process has two main parts: ED and UC. In an ED problem, all the units are assumed to be already connected to the grid. This is the problem that we have been investigating so far in this paper. Closely associated with the ED problem is the UC problem, where the units that are to be connected to the power grid are selected. This section mainly discusses the incorporation of versatile distribution model into UC problems.

Compared to ED, the on/off status of each unit should be additionally determined in UC problems. The UC model including wind power can hence be built from the extension of the ED model used in this paper (see Section A of the Appendix). The classical solution methods of UC consist of priority-list method, dynamic-programming solution, Lagrange relaxation solution, etc. [15]. The procedures of these methods involve the ED problem as a subproblem. That is, for each subset of the units that are to be determined as on or off, for any given set of them connected to the grid, the particular subset should be operated in optimum economic fashion [15]. Since the application of versatile distribution can improve the accuracy and speed of ED computation, it can surely improve UC.

It becomes popular these years to solve UC problems by scenario generation approaches. This approach describes uncertain wind power with a set of scenarios and each scenario contains a deterministic power and its probability. Section B of the Appendix shows a UC model on the basis of introducing wind power uncertainties, which are represented in the form of scenarios. The scenarios are always generated through sampling from an inverse CDF and Gaussian distribution was mostly used so far [30]. Now, the scenarios can be sampled from the inverse CDF of versatile distribution, which is obviously faster and more accurate. It should be noted that scenario generation is time-consuming once a large number of scenarios must be sampled to meet the requirement of accuracy [22]. Therefore, it can only be adopted when the requirement of computation speed is not so strict.

VI. Conclusion

The theoretical distributions such as Gaussian and Beta are sometimes not reasonable to represent wind power forecast errors. Moreover, the use of these distributions raises mathematical challenges for the ED problem incorporating wind power to be solved by the classical SLP-based algorithm. Under this background, this paper presents a versatile distribution model which can well represent the distribution of wind power forecast errors for all forecast timescales and magnitudes. The results of qualitative and quantitative analysis show that the versatile distribution exhibits a better matching for the actual distribution of forecast errors compared with Gaussian and Beta distributions. The incorporation of versatile distribution can also simplify the ED problems with wind power. As a result, the ED problems can be solved by the SLP-based algorithm again, which provides an assistive tool for system operators in future. The case study results show that the proposed SLP-based algorithm using the versatile distribution is not only the most precise but also the fastest compared with the cases of using Gaussian and Beta distributions. Since ED is always a subproblem of a UC problem, the incorporation of versatile distribution can also enhance the computation efficiency for UC problems.

Appendix

A. UC Model Incorporating the Versatile Distribution Model

The UC model incorporating the versatile distribution model can be formulated as follows [30]–[32]:

$$\begin{aligned}\min \sum_{i=1}^{I}\sum_{t=1}^{T}&[C_i(p_{i,t}) + SU_{i,t} + SD_{i,t}]\\ &+\sum_{j=1}^{J}\sum_{t=1}^{T}[C_{w,j}(w_{j,t}) + C_{p,j}(w_{av,j,t} - w_{j,t})\\ &\qquad +C_{r,j}(w_{j,t} - w_{av,j,t})]\end{aligned} \tag{30}$$

$$\text{s.t.} \sum_{i=1}^{I} p_{i,t} + \sum_{j=1}^{J} w_{j,t} = L_t, \forall t \tag{31}$$

$$\eta_{i,t}p_{\min,i} \le p_{i,t} \le \eta_{i,t}p_{\max,i}, \forall i, \forall t \tag{32}$$

$$0 \le w_{j,t} \le w_{r,j}, \forall j, \forall t \tag{33}$$

$$0 \le r_{u,i,t} \le \min\{\eta_{i,t}p_{\max,i} - p_{i,t}, \eta_{i,t}r_{u,\max,i}\}, \forall i, \forall t \tag{34}$$

$$0 \le r_{d,i,t} \le \min\{p_{i,t} - \eta_{i,t}p_{\min,i}, \eta_{i,t}r_{d,\max,i}\}, \forall i, \forall t \tag{35}$$

$$\sum_{j=1}^{J} w_{j,t} - \sum_{i=1}^{I} r_{u,i,t} \le F_{\Sigma,t}^{-1}(1 - c_u), \forall t \tag{36}$$

$$\sum_{j=1}^{J} w_{j,t} + \sum_{i=1}^{I} r_{d,i,t} \ge F_{\Sigma,t}^{-1}(c_d), \forall t \tag{37}$$

$$-\mathbf{F}^{\max} \le \mathbf{F}_t \le \mathbf{F}^{\max}, \forall t \tag{38}$$

$$\begin{aligned}p_{i,t} - p_{i,t-1} \le\ & \eta_{i,t-1}R_{u,\max,i} + (\eta_{i,t} - \eta_{i,t-1})p_{\min,i}\\ &+ (1 - \eta_{i,t})p_{\max,i}, \forall i, \forall t\end{aligned} \tag{39}$$

$$\begin{aligned}p_{i,t-1} - p_{i,t} \le\ & \eta_{i,t}R_{d,\max,i} + (\eta_{i,t-1} - \eta_{i,t})p_{\min,i}\\ &+ (1 - \eta_{i,t-1})p_{\max,i}, \forall i, \forall t\end{aligned} \tag{40}$$

$$(t_{i,on,t} - T_{i,on})(\eta_{i,t-1} - \eta_{i,t}) \ge 0, \forall i, \forall t \tag{41}$$

$$(t_{i,off,t} - T_{i,off})(\eta_{i,t} - \eta_{i,t-1}) \ge 0, \forall i, \forall t \tag{42}$$

$$SU_{i,t} \ge \max\{K_i(\eta_{i,t} - \eta_{i,t-1}), 0\}, \forall i, \forall t \tag{43}$$

$$SD_{i,t} \ge \max\{M_i(\eta_{i,t-1} - \eta_{i,t}), 0\}, \forall i, \forall t \tag{44}$$

where t is the index of time period, $t = 1, \ldots, T$, $SU_{i,t}$ is the startup cost of unit i at time t, $SD_{i,t}$ is the shutdown cost of unit i at time t, $\eta_{i,t}$ is the binary on/of variable of unit i at time t, $\mathbf{F}_t$ is the vector of branch flow at time t, $\mathbf{F}^{\max}$ is the vector of transmission constraints of branches, $t_{i,on,t}$ is the on time of unit i up to t, $t_{i,off,t}$ is the off time of unit i up to t, $T_{i,on}$ is the minimum on time of unit i, $T_{i,off}$ is the minimum off time of unit i, K_i is the startup price of unit i, and M_i is the shutdown price of unit i.

B. UC Model Based on Scenario Generation Approach

The UC model where wind power uncertainties are represented by scenarios can be formulated as follows [30]–[32]:

$$\min \sum_{s=1}^{S} prob^{(s)} \cdot \left\{ \sum_{i=1}^{I}\sum_{t=1}^{T}\left[C_i\left(p_{i,t}^{(s)}\right)+SU_{i,t}^{(s)}+SD_{i,t}^{(s)}\right] + \sum_{j=1}^{J}\sum_{t=1}^{T}\left[C_{w,j}\left(w_{j,t}^{(s)}\right)+C_{p,j}\left(w_{av,j,t}^{(s)}-w_{j,t}^{(s)}\right) + C_{r,j}\left(w_{j,t}^{(s)}-w_{av,j,t}^{(s)}\right)\right]\right\} \tag{45}$$

s.t.

$$\sum_{i=1}^{I} p_{i,t}^{(s)}+\sum_{j=1}^{J} w_{j,t}^{(s)}=L_t, \forall t, \forall s \tag{46}$$

$$\eta_{i,t}^{(s)} p_{\min,i} \leq p_{i,t}^{(s)} \leq \eta_{i,t}^{(s)} p_{\max,i}, \forall i, \forall t, \forall s \tag{47}$$

$$0 \leq w_{j,t}^{(s)} \leq w_{r,j}, \forall j, \forall t, \forall s \tag{48}$$

$$0 \leq r_{u,i,t}^{(s)} \leq \min\left\{\eta_{i,t}^{(s)} p_{\max,i}-p_{i,t}^{(s)}, \eta_{i,t}^{(s)} r_{u,\max,i}\right\}, \forall i, \forall t, \forall s \tag{49}$$

$$0 \leq r_{d,i,t}^{(s)} \leq \min\left\{p_{i,t}^{(s)}-\eta_{i,t}^{(s)} p_{\min,i}, \eta_{i,t}^{(s)} r_{d,\max,i}\right\}, \forall i, \forall t, \forall s \tag{50}$$

$$\sum_{i=1}^{I} r_{u,i,t}^{(s)} \geq \sum_{j=1}^{J}\left(w_{j,t}^{(s)}-w_{av,j,t}^{(s)}\right), \forall t, \forall s \tag{51}$$

$$\sum_{i=1}^{I} r_{d,i,t}^{(s)} \geq \sum_{j=1}^{J}\left(w_{av,j,t}^{(s)}-w_{j,t}^{(s)}\right), \forall t, \forall s \tag{52}$$

$$-\mathbf{F}^{\max} \leq \mathbf{F}_t^{(s)} \leq \mathbf{F}^{\max}, \forall t, \forall s \tag{53}$$

$$p_{i,t}^{(s)}-p_{i,t-1}^{(s)} \leq \eta_{i,t-1}^{(s)} R_{u,\max,i}+\left(\eta_{i,t}^{(s)}-\eta_{i,t-1}^{(s)}\right) p_{\min,i} + \left(1-\eta_{i,t}^{(s)}\right) p_{\max,i}, \forall i, \forall t, \forall s \tag{54}$$

$$p_{i,t-1}^{(s)}-p_{i,t}^{(s)} \leq \eta_{i,t}^{(s)} R_{d,\max,i}+\left(\eta_{i,t-1}^{(s)}-\eta_{i,t}^{(s)}\right) p_{\min,i} + \left(1-\eta_{i,t-1}^{(s)}\right) p_{\max,i}, \forall i, \forall t, \forall s \tag{55}$$

$$\left(t_{i,on,t}^{(s)}-T_{i,on}\right)\left(\eta_{i,t-1}^{(s)}-\eta_{i,t}^{(s)}\right) \geq 0, \forall i, \forall t, \forall s \tag{56}$$

$$\left(t_{i,off,t}^{(s)}-T_{i,off}\right)\left(\eta_{i,t}^{(s)}-\eta_{i,t-1}^{(s)}\right) \geq 0, \forall i, \forall t, \forall s \tag{57}$$

$$SU_{i,t}^{(s)} \geq \max\left\{K_i\left(\eta_{i,t}^{(s)}-\eta_{i,t-1}^{(s)}\right), 0\right\}, \forall i, \forall t, \forall s \tag{58}$$

$$SD_{i,t}^{(s)} \geq \max\left\{M_i\left(\eta_{i,t-1}^{(s)}-\eta_{i,t}^{(s)}\right), 0\right\}, \forall i, \forall t, \forall s \tag{59}$$

where s is the index for scenario, $s=1,\ldots,S, prob^{(s)}$ is the probability of scenario s.

Note that the available wind power $w_{av,j,t}^{(s)}$ (deterministic variable) is obtained through sampling from the inverse CDF of the versatile distribution

REFERENCES

[1] T. Ackermann, *Wind Power in Power System*. London, U.K.: Wiley, 2005.

[2] G. Giebel, R. Brownsword, G. Kariniotakis, M. Denhard, and C. Draxl, The State-of-the-Art in Short-Term Prediction of Wind Power: A Literature Overview, Deliverable D-1.2 of ANEMOS project, Jan. 2011. [Online]. Available: http://130.226.56.153/zephyr/publ/GGiebelEtAl-StateOfTheArtInShortTermPrediction_ANEMOSplus_2011.pdf.

[3] R. Doherty and M. O'Malley, "A new approach to quantify reserve demand in systems with significant installed wind capacity," *IEEE Trans. Power Syst.*, vol. 20, no. 2, pp. 587–595, May 2005.

[4] F. Bouffard and F. D. Galiana, "Stochastic security for operations planning with significant wind power generation," *IEEE Trans. Power Syst.*, vol. 23, no. 2, pp. 306–316, May 2008.

[5] Z.-S. Zhang, Y.-Z. Sun, G.-J. Li, L. Cheng, and J. Lin, "A solution of the economic dispatch problem considering wind power uncertainty," *Autom. Elect. Power Syst.*, vol. 35, no. 22, pp. 125–130, Nov. 2011.

[6] S. Bofinger, A. Luig, and H. G. Beyer, "Qualification of wind power forecasts," in *Proc. Global Wind Power Conf.*, Paris, France, 2002.

[7] A. Fabbri, T. G. S. Roman, J. R. Abbad, and V. H. M Quezada, "Assessment of the cost associated with wind generation prediction errors in a liberalized electricity market," *IEEE Trans. Power Syst.*, vol. 20, no. 3, pp. 1440–1446, Aug. 2005.

[8] B.-M. Hodge and M. Milligan, "Wind power forecasting error distributions over multiple timescales," in *Proc. IEEE Power and Energy Society General Meeting*, Detroit, MI, USA, 2011.

[9] M. Lange, "On the uncertainty of wind power predictions—Analysis of the forecast accuracy and statistical distribution of errors," *J. Solar Energy Eng.*, vol. 127, pp. 177–184, May 2005.

[10] R. A. Jabr and B. C. Pal, "Intermittent wind generation in optimal power flow dispatching," *IET Gener., Transm., Distrib.*, vol. 3, no. 1, pp. 66–74, 2009.

[11] V. S. Pappala, I. Erlich, K. Rohrig, and J. Dobschinski, "A stochastic model for the optimal operation of a wind-thermal power system," *IEEE Trans. Power Syst.*, vol. 24, no. 2, pp. 940–950, May 2009.

[12] V. Miranda and P. S. Hang, "Economic dispatch model with fuzzy wind constraints and attitudes of dispatchers," *IEEE Trans. Power Syst.*, vol. 20, no. 4, pp. 2143–2145, Nov. 2005.

[13] J. Hetzer, D. C. Yu, and K. Bhattarai, "An economic dispatch model incorporating wind power," *IEEE Trans. Energy Convers.*, vol. 23, no. 2, pp. 603–611, Jun. 2008.

[14] M. H. Albadi and E. F. El-Saadany, "Comparative study on impacts of wind profiles on thermal units scheduling costs," *IET Renew. Power Gener.*, vol. 5, no. 1, pp. 26–35, 2011.

[15] A. J. Wood and B. F. Wollenberg, *Power Generation, Operation, Control*, 2nd ed. Beijing, China: Tsinghua Univ. Press, 2003.

[16] X. Yan and V. H. Quintana, "An efficient predictor-corrector interior point algorithm for security-constrained economic dispatch," *IEEE Trans. Power Syst.*, vol. 12, no. 2, pp. 803–810, May 1997.

[17] H. B. Gooi, D. P. Mendes, K. R. W. Bel, and D. S. Kirschen, "Optimal scheduling of spinning reserve," *IEEE Trans. Power Syst.*, vol. 14, no. 4, pp. 1485–1492, Nov. 1999.

[18] L. Wang and C. Singh, "Balancing risk and cost in fuzzy economic dispatch including wind power penetration based on particle swarm optimization," *Elect. Power Syst. Res.*, vol. 78, no. 8, pp. 1361–1368, Aug. 2008.

[19] Y.-Y. Hong and C.-T. Li, "Short-term real-power scheduling considering fuzzy factors in an autonomous system using genetic algorithms," *Proc. Inst. Elect. Eng., Gener., Transm., Distrib.*, vol. 153, no. 6, pp. 684–692, Nov. 2006.

[20] C.-L. Chen, "Simulated annealing-based optimal wind-thermal coordination scheduling," *IET Gener., Transm., Distrib.*, vol. 1, no. 3, pp. 447–455, 2007.

[21] T.-Y. Lee, "Optimal spinning reserve for a wind-thermal power system using EIPSO," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 1612–1621, Nov. 2007.

[22] Z.-S. Zhang, Y.-Z. Sun, J. Lin, L. Cheng, and G.-J. Li, "Versatile distribution of wind power output for a given forecast value," in *Proc. 2012 IEEE Power & Energy Society General Meeting*, San Diego, CA, USA, Jul. 22–26, 2012.

[23] U. A. Ozturk, M. Mazumdar, and B. A. Norman, "A solution to the stochastic unit commitment problem using chance constrained programming," *IEEE Trans. Power Syst.*, vol. 19, no. 3, pp. 1589–1598, Aug. 2004.

[24] A. T. Al-Awami and M. A. El-Sharkawi, "Statistical characterization of wind power output for a given wind power forecast," in *Proc. North American Power Symp.*, Starkville, MS, USA, 2009.

[25] Z.-S. Zhang, Y.-Z. Sun, J. Lin, V. Modi, L. Cheng, and G. Chen, "Innovation for wind power: Adding smart processors between prediction systems and dispatch systems," in *Proc. IEEE PES Transmission and Distribution Conf. Expo.*, Orlando, FL, USA, May 2012.

[26] H. Bludszuweit, J. A. Domínguez-Navarro, and A. Llombart, "Statistical analysis of wind power forecast error," *IEEE Trans. Power Syst.*, vol. 23, no. 3, pp. 983–991, Aug. 2008.

[27] J. Liang, S. Grijalva, and R. G. Harley, "Increased wind revenue and system security by trading wind power in energy and regulation reserve markets," *IEEE Trans. Sustain. Energy*, vol. 2, no. 3, pp. 340–347, Jul. 2011.

[28] C.-M. Huang, H.-T. Yang, and C.-L. Huang, "Bi-objective power dispatch using fuzzy satisfaction-maximizing decision approach," *IEEE Trans. Power Syst.*, vol. 12, no. 4, pp. 1715–1721, Nov. 1997.

[29] S. Mishra, Y. Mishra, and S. Vignesh, "Security constrained economic dispatch considering wind energy conversion systems," in *Proc. IEEE Power and Energy Society General Meeting*, Detroit, MI, USA, 2011.

[30] J. Wang, A. Botterud, R. Bessa, H. Keko, L. Carvalho, D. Issicaba, J. Sumaili, and V. Miranda, "Wind power forecasting uncertainty and unit commitment," *Appl. Energy*, vol. 88, pp. 4014–4023, 2011.

[31] Y. Wang, Q. Xia, and C. Kang, "Unit commitment with volatile node injections by using interval optimization," *IEEE Trans. Power Syst.*, vol. 26, no. 3, pp. 1705–1713, Aug. 2011.

[32] J. Wang, M. Shahidehpour, and Z. Li, "Security-constrained unit commitment with volatile wind power generation," *IEEE Trans. Power Syst.*, vol. 23, no. 3, pp. 1319–1327, Aug. 2008.

Zhao-Sui Zhang (S'12) received the B.S. degree in electrical engineering from Tsinghua University, Beijing, China, in 2009, where he is pursuing the Ph.D. degree in electrical engineering.

He is now sponsored by China Scholarship Council (CSC) to be a joint Ph.D. student at Columbia University, New York, NY, USA. His main research interests are wind power, power/frequency regulation, power economics, electricity markets, and energy policy.

Yuan-Zhang Sun (M'99–SM'01) received the B.S. degree from Wuhan University of Hydro and Electrical Engineering, China, the M.S. degree from Electric Power Research Institute (EPRI), China, and Ph.D. degree from Tsinghua University, Beijing, China, in 1978, 1982, and 1988, respectively, all in electrical engineering.

He is now a chair professor at Tsinghua University and dean of the School of Electrical Engineering at Wuhan University. His main research interests are in the areas of power systems dynamics and control, voltage stability and control, and renewable energy.

David Wenzhong Gao (M'02–SM'03) received the M.S. and Ph.D. degrees in electrical and computer engineering specializing in electric power engineering from Georgia Institute of Technology, Atlanta, GA, USA, in 1999 and 2002, respectively.

He is now an associate professor at the Department of Electrical and Computer Engineering, University of Denver, Denver, CO, USA. His current teaching and research interests include renewable energy and distributed generation, smart grid, power system protection, power electronics applications in power systems, power system modeling and simulation, and hybrid electric propulsion systems.

Dr. Gao is an Editor for the IEEE TRANSACTIONS ON SUSTAINABLE ENERGY. He is the General Chair for The IEEE Symposium on Power Electronics and Machines in Wind Applications (PEMWA 2012).

Jin Lin (S'11) received the B.S. degree and Ph.D. degree in electrical engineering from Tsinghua University, Beijing, China, in 2007 and 2012, respectively.

He is now a Postdoctoral Fellow at Tsinghua University. His main research interests are renewable energy and the stability analysis on power systems.

Lin Cheng (M'05) received the B.S. degree from Tianjin University in 1996 and the Ph.D. degree from Tsinghua University, Beijing, China, in 2001.

Currently, he is an associate professor of the Electrical Engineering Department at Tsinghua University. His main research interests are in the areas of power system reliability, power system dynamics and control, voltage stability, and control.